

# Kalman Filtering 101: A Step-by-Step Derivation from the Ground Up

Control Systems Engineering Primer

July 22, 2026

## Abstract

This document provides a rigorous, step-by-step mathematical derivation of the discrete-time linear Kalman filter. By formulating the estimator as a recursive linear predictor-corrector and applying optimization via trace minimization of the error covariance, we bridge the gap between stochastic derivation and deterministic optimal estimation.

## 1 System Modeling and Definitions

Consider a linear, time-invariant discrete-time system governed by the state-space equations:

$$x_k = Ax_{k-1} + B_u u_{k-1} + w_{k-1} \quad (1)$$

$$y_k = Cx_k + v_k \quad (2)$$

where:

- $x_k \in \mathbb{R}^n$  is the true state vector at time step  $k$ .
- $u_k \in \mathbb{R}^m$  is the known deterministic input vector.
- $y_k \in \mathbb{R}^p$  is the sensor measurement vector.
- $w_k \in \mathbb{R}^n$  is the process noise (or disturbance) with zero mean and covariance matrix  $Q = E[w_k w_k^T]$ .
- $v_k \in \mathbb{R}^p$  is the measurement noise with zero mean and covariance matrix  $R = E[v_k v_k^T]$ , independent of  $w_k$ .

## 2 The Predict-Correct Structure

The Kalman filter operates recursively in two distinct phases:

### 2.1 1. Prediction (Time Update)

Before receiving the new sensor measurement at step  $k$ , we project our state estimate and error covariance forward using the system model:

$$\hat{x}_{k|k-1} = A\hat{x}_{k-1|k-1} + B_u u_{k-1} \quad (3)$$

$$P_{k|k-1} = AP_{k-1|k-1}A^T + Q \quad (4)$$

where  $\hat{x}^-$  denotes the *a priori* estimate and  $P^-$  is the *a priori* error covariance.

## 2.2 2. Correction (Measurement Update)

Once the measurement  $y_k$  arrives, we correct our prediction using a linear blending gain  $K_k$  (the Kalman gain):

$$\hat{x}_{k|k} = \hat{x}_{k|k-1} + K_k (y_k - C\hat{x}_{k|k-1}) \quad (5)$$

The term  $\tilde{y}_k = y_k - C\hat{x}_{k|k-1}$  is defined as the **innovation** (or measurement residual).

## 3 Rigorous Derivation of the Error Covariance ( $P$ )

To find the optimal gain  $K_k$ , we must first determine how the estimation error propagates through the measurement update.

### 3.1 Step A: Defining the Error and the Joseph Form

The *a priori* and *a posteriori* estimation errors are defined as:

$$e_{k|k-1} = x_k - \hat{x}_{k|k-1} \quad (6)$$

$$e_{k|k} = x_k - \hat{x}_{k|k} \quad (7)$$

Substituting the correction equation and the measurement equation ( $y_k = Cx_k + v_k$ ) into the *a posteriori* error yields:

$$e_{k|k} = x_k - [\hat{x}_{k|k-1} + K_k (Cx_k + v_k - C\hat{x}_{k|k-1})] \quad (8)$$

$$= (x_k - \hat{x}_{k|k-1}) - K_k C(x_k - \hat{x}_{k|k-1}) - K_k v_k \quad (9)$$

$$= (I - K_k C)e_{k|k-1} - K_k v_k \quad (10)$$

The *a posteriori* error covariance matrix  $P_{k|k} = E[e_{k|k}e_{k|k}^T]$  is then calculated by taking the expectation. Because process/measurement noise and estimation errors are mutually uncorrelated ( $E[e_{k|k-1}v_k^T] = 0$ ), the cross-terms vanish:

$$P_{k|k} = (I - K_k C)P_{k|k-1}(I - K_k C)^T + K_k R K_k^T \quad (11)$$

This expression is known as the **Joseph form** of the covariance update, which holds for any arbitrary gain  $K_k$ .

### 3.2 Step B: Expanding the Covariance Matrix

Expanding equation (10) algebraically:

$$P_{k|k} = P_{k|k-1} - K_k C P_{k|k-1} - P_{k|k-1} C^T K_k^T + K_k (C P_{k|k-1} C^T + R) K_k^T \quad (12)$$

## 4 Deriving the Optimal Kalman Gain

To find the gain  $K_k$  that minimizes the total estimation uncertainty, we minimize the sum of the variances on the main diagonal—represented mathematically by the trace of  $P_{k|k}$ :

$$\min_{K_k} \text{Tr}(P_{k|k}) \quad (13)$$

Using standard matrix calculus identities ( $\frac{\partial \text{Tr}(AB)}{\partial A} = B^T$  and  $\frac{\partial \text{Tr}(ACA^T)}{\partial A} = 2AC$  for symmetric  $C$ ), we take the derivative of  $\text{Tr}(P_{k|k})$  with respect to  $K_k$  and set it to zero:

$$\frac{\partial \text{Tr}(P_{k|k})}{\partial K_k} = -2 (P_{k|k-1} C^T)^T + 2K_k (C P_{k|k-1} C^T + R) = 0 \quad (14)$$

Solving for  $K_k$ :

$$K_k (CP_{k|k-1}C^T + R) = P_{k|k-1}C^T \quad (15)$$

$$\boxed{K_k = P_{k|k-1}C^T (CP_{k|k-1}C^T + R)^{-1}} \quad (16)$$

Letting  $S_k = CP_{k|k-1}C^T + R$  represent the innovation covariance, the optimal gain is written compactly as  $K_k = P_{k|k-1}C^T S_k^{-1}$ .

## 5 Simplifying the Covariance Update

Substituting the optimal gain condition  $P_{k|k-1}C^T = K_k S_k$  back into the expanded covariance equation (11) yields a massive algebraic simplification:

$$P_{k|k} = P_{k|k-1} - K_k C P_{k|k-1} - \cancel{P_{k|k-1}C^T K_k^T} + \cancel{K_k S_k K_k^T} \quad (17)$$

$$= (I - K_k C) P_{k|k-1} \quad (18)$$

This proves the standard, highly efficient computational form for the updated error covariance.